Beyond the Lexicon: A Comparative Evaluation of VADER and GoEmotions on Music Lyrics

Sentiment Analysis, Affective Classification, and the Limits of Lexical Approaches

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Abstract

Lexical and neural sentiment analysis tools have demonstrated strong performance on social media text and product reviews, but their effectiveness on music lyrics remains underexplored. This study conducts a comparative evaluation of two sentiment analysis systems — VADER (Valence Aware Dictionary and sEntiment Reasoner), a rule-based lexical tool, and GoEmotions, a neural multi-label emotion classifier — applied to a curated dataset of 25 popular songs spanning nine emotionally distinct categories, including Romantic, Longing, Nostalgic, Bittersweet, Empowering, Intense, and Tragic. Computed scores and predicted emotion labels are compared against human-assigned emotional classifications and intensity ratings. VADER achieved binary concordance in only 32.0% of cases, with emotionally complex categories spanning the near-full sentiment range (–0.97 to +0.99). GoEmotions achieved substantially higher weighted concordance of 66.0% (full match: 48.0%), demonstrating superior sensitivity to nuanced affective registers but exhibiting its own systematic failure modes including emotional fragmentation and dominant neutral prediction. Qualitative analysis identifies four VADER failure types — metaphor blindness, valence inversion, affective collapse, and accidental accuracy — and three GoEmotions failure modes: affective ambiguity inflation, genre blindness, and emotional granularity collapse. These findings indicate that both systems fall short of human affective classification on lyrical text, though neural multi-label approaches represent a meaningful advance. Results point toward the need for multimodal or lyric-specific context-aware models for robust music sentiment analysis.

Keywords: sentiment analysis, music lyrics, VADER, GoEmotions, affective computing, music information retrieval, natural language processing

1. Introduction

Music occupies a unique position in human emotional life. A song categorized as lyrically "negative" — filled with words of absence, longing, or loss — may nonetheless produce feelings of catharsis, empowerment, or profound connection in its listener. This gap between lexical content and affective experience is intuitively familiar to anyone who has found comfort in a minor key, or felt strangely uplifted by a song about grief. It is precisely this gap that makes music lyrics a demanding and revealing test case for sentiment analysis systems.

Natural language processing has made significant advances in affective computing, with tools ranging from rule-based lexical analyzers to large pretrained neural classifiers achieving strong performance on short-form social media text and product reviews (Hutto & Gilbert, 2014; Demszky et al., 2020). These tools operate on fundamentally different principles: lexical approaches such as VADER assign sentiment scores based on word-level valence dictionaries, while neural approaches such as GoEmotions use transformer-based models fine-tuned on large annotated corpora to predict fine-grained emotion labels. Music lyrics, however, present a domain that strains both paradigms. Lyrics rely heavily on metaphor, irony, prosodic ambiguity, intertextual reference, and cultural context to convey emotional meaning that frequently inverts or complicates surface-level word choice (Fell & Sporleder, 2014).

This study investigates these limitations comparatively. Using a dataset of 25 popular songs spanning nine emotional categories, we apply both VADER sentiment analysis and GoEmotions multi-label emotion classification and compare results against human-assigned emotional categories and intensity ratings. We find that VADER achieves binary concordance in only 32.0% of cases and GoEmotions achieves a weighted concordance rate of 66.0%, and we identify systematic failure modes that illuminate the structural and architectural limitations of each approach when applied to musical language. These findings have practical implications for music information retrieval (MIR), playlist curation, and the development of emotionally intelligent recommendation systems.

2. Related Work

2.1 Sentiment Analysis Tools

Hutto and Gilbert (2014) introduced VADER as a rule-based sentiment analysis tool optimized for social media text, demonstrating high accuracy on Twitter data and product reviews. VADER assigns valence scores to a lexicon of over 7,500 features and computes a compound score normalized between –1 and +1, incorporating heuristics for capitalization, punctuation, and degree modifiers. While robust on direct emotional expression, the authors acknowledge limitations with figurative or ironic language — a limitation directly relevant to the lyrical domain.

Demszky et al. (2020) introduced GoEmotions, a dataset of 58,000 Reddit comments annotated for 28 fine-grained emotion categories, and accompanying neural classifiers achieving state-of-the-art performance on multi-label emotion recognition. GoEmotions represents a paradigm shift from scalar sentiment to categorical emotion: rather than producing a single positivity score, it predicts the probability of each of 28 emotions and a neutral label, enabling richer characterization of complex affective states. However, GoEmotions was trained on Reddit data — conversational, predominantly English, and stylistically distinct from poetic or musical language.

2.2 Music, Emotion, and Computation

The relationship between musical features and perceived emotion has been studied extensively in music information retrieval. Kim et al. (2010) demonstrated that acoustic features — tempo, mode, loudness, and spectral characteristics — correlate strongly with listener-reported emotion, often more reliably than lyrical content alone. This finding has a crucial implication: any purely textual approach to music sentiment analysis is structurally incomplete, regardless of its performance on other text types.

Fell and Sporleder (2014) applied computational methods specifically to lyrics classification, finding that lyrics encode genre and mood but that bag-of-words approaches struggle substantially with figurative language. Their work highlights the particular challenge of songs that employ negative-valence vocabulary to express positive or ambivalent emotional states — a pattern that emerges repeatedly in the present dataset. More recently, Çano and Morisio (2017) demonstrated that combining acoustic and lyrical features substantially outperforms either modality alone, reinforcing the multimodal direction for future work.

In the broader domain of narrative text, Reagan et al. (2016) identified six core emotional arc shapes using sentiment analysis over narrative time, finding that emotionally complex arcs are most valued by readers. A parallel hypothesis applies to music: songs with emotionally complex or contradictory sentiment profiles may be precisely those most meaningful to listeners — and most challenging for automated systems.

Acheampong et al. (2020) conducted a comprehensive review of text-based emotion detection, identifying key limitations of lexicon-based and neural approaches including cultural specificity, ambiguity handling, and cross-domain generalization — all of which bear directly on the present study.

3. Methodology

3.1 Dataset

A dataset of 25 popular songs was compiled across nine emotional categories: Romantic (n=8), Longing (n=3), Nostalgic (n=3), Bittersweet (n=3), Reflective (n=2), Empowering (n=1), Comforting (n=1), Tragic (n=1), and Intense (n=3). Songs were selected to represent a range of emotional registers within each category, with particular attention to cases where surface lyrical content might diverge from perceived emotional experience — precisely the conditions under which automated systems are most likely to fail.

Emotional categories and intensity ratings (1–10) were assigned by the author based on personal listening experience and general cultural familiarity with each track. The author acknowledges this as a form of expert annotation rather than population-representative labeling; inter-rater reliability was not computed for this study, which is noted as a limitation. Categories were assigned using standard affective terminology consistent with prior music emotion research (Russell, 1980; Kim et

al., 2010).

3.2 VADER Sentiment Analysis

Lyrics were retrieved using the LyricsGenius Python library via the Genius API. VADER compound sentiment scores were computed for each song's full lyrical text using the vaderSentiment library. Compound scores range from –1 (most negative) to +1 (most positive). Match classification was performed by comparing the directional valence of the VADER compound score (positive/negative) against the assigned emotional category. Songs in inherently ambiguous categories (Bittersweet, Reflective) were evaluated on whether the score reflected their tonal complexity. Binary concordance was assigned (1 = match, 0 = mismatch) with no partial credit.

3.3 GoEmotions Classification

GoEmotions multi-label emotion classification was applied to each song's lyrical text using the pretrained model fine-tuned on 28 emotion categories plus neutral (Demszky et al., 2020). The top five predicted emotion labels by probability were recorded per song. Match scoring used a graded system reflecting the richer output format: full match (score = 1.0) where the top predicted emotions directly corresponded to the assigned category; mostly match (score = 0.75) where dominant predictions partially aligned; partly match (score = 0.5) where relevant emotions appeared but were not dominant; and no match (score = 0) where predicted emotions showed no meaningful correspondence to the assigned category. This graded scoring was applied because GoEmotions produces ranked probability distributions rather than binary outputs, making binary scoring alone an inappropriate comparator.

3.4 Comparative Analysis Framework

Both systems were evaluated against identical human-assigned ground truth labels. For VADER, the primary metric is binary concordance rate. For GoEmotions, both full match rate and weighted concordance rate (mean of graded scores × 100) are reported. Qualitative failure analysis was conducted across both systems to identify recurring error patterns, which are synthesized into taxonomic failure modes in Section 5.

4. Results

4.1 VADER Performance

VADER produced binary-concordant classifications for 8 of 25 songs (32.0%), yielding a mismatch rate of 68.0%. This result indicates substantial disagreement between VADER classifications and human-assigned emotional categories. VADER compound scores exhibited a strongly bimodal distribution: 4 songs clustered at extreme negative values (–0.91 to –0.97) and 17 songs clustered at high positive values (+0.80 to +0.998). Only 4 songs produced mid-range scores (Golden Brown: 0.052; Notion: 0.345; End of Beginning: 0.506; Toxic: 0.001). This near-total bimodality indicates severely limited sensitivity to the emotional middle ground that characterizes much of popular music.

Category-level analysis reveals that emotionally complex categories produce the highest within-category VADER score variance. The Romantic category spans a range of 1.949 (min: –0.9515, max: +0.9977), and the Longing category spans 1.961 (min: –0.9709, max: +0.9904) — effectively the full possible sentiment range. This suggests limited predictive validity for that affective dimension. The Intense category spans 0.997 (0.001 to 0.998), reflecting a similar collapse. By contrast, the Bittersweet category produced a range of only 0.124 (0.798 to 0.922), suggesting that VADER paradoxically collapses emotionally ambiguous content into near-identical high-positive scores.

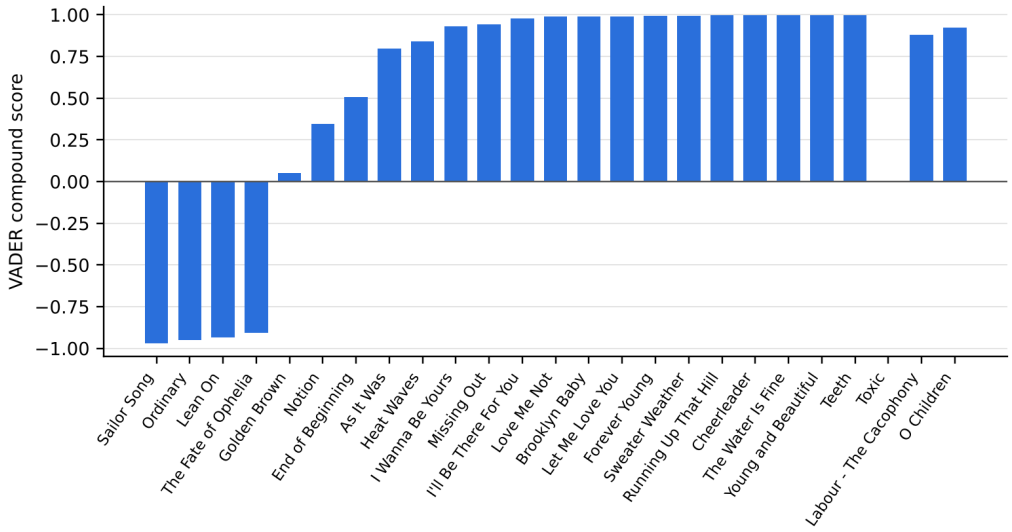


Figure 1: VADER Scores of Songs

Songs rated highest in emotional intensity — Forever Young (Alphaville, 10), Running Up That Hill (Kate Bush, 10), and Labour: The Cacophony (Paris Paloma, 10) — all produced classification mismatches, as did Love Me Not (Ravyn Lenae, intensity 9) and The Water Is Fine (Chloe Ament, intensity 9). The two comfortably correct matches at high intensity were I'll Be There For You (The Rembrandts, 9) and Sweater Weather (The Neighbourhood, 9) — both songs whose lyrical

positivity is relatively explicit. This suggests VADER's accuracy at high intensity is conditional on lexical directness.

4.2 GoEmotions Performance

GoEmotions achieved full concordance for 12 of 25 songs (48.0%), mostly-match for 4 songs (16.0%), partly-match for 3 songs (12.0%), and no match for 6 songs (24.0%). The weighted concordance rate — calculated as the mean of graded match scores — was 66.0%, representing a substantial improvement over VADER's 32.0% binary rate. When compared on an equivalent binary basis (full match only), GoEmotions' 48.0% still meaningfully exceeds VADER's 32.0%.

GoEmotions demonstrated particular strengths in categories involving desire, sadness, and interpersonal connection. The Longing category, where VADER failed on 2 of 3 songs, saw GoEmotions achieve full or mostly-match on 2 of 3 cases through detection of desire and love as dominant emotions. The Nostalgic category saw GoEmotions achieve full match on 2 of 3 songs, correctly identifying sadness, disappointment, and neutral as characteristic of retrospective affect. The Comforting category (I'll Be There For You) was correctly identified through caring, joy, and optimism labels.

GoEmotions performed weakest on the Intense and Empowering categories. Running Up That Hill (Empowering) received a no-match, with neutral and curiosity dominating — an output that captures neither the song's lyrical struggle nor its affective register. Teeth (Intense) received no match, with love and admiration as dominant labels — a striking inversion driven by the song's intimate second-person address. Labour: The Cacophony (Intense) also received no match, reflecting the model's difficulty with anger and defiance expressed through complexity and ambiguity rather than explicit vocabulary.

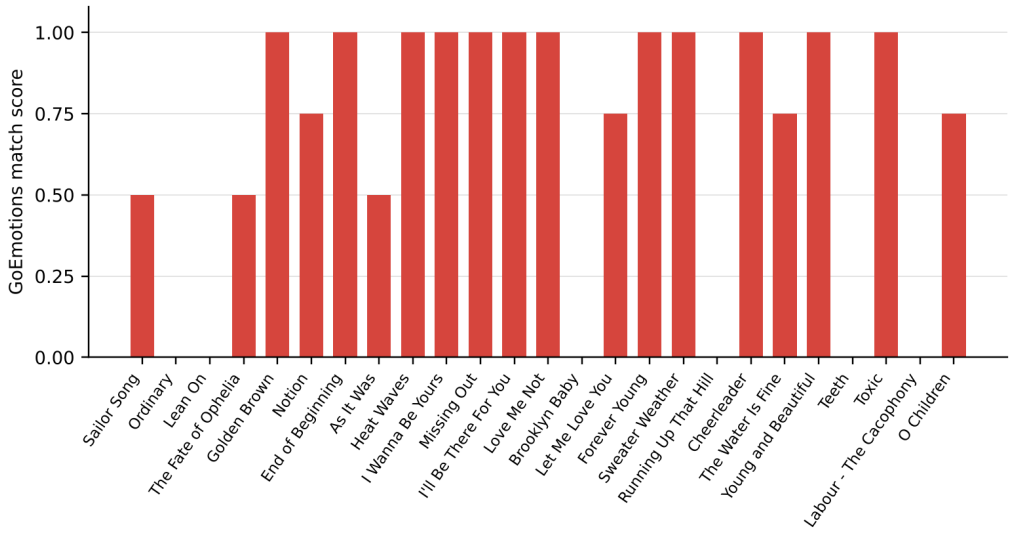


Figure 2: GoEmotions Match Scores of Songs

4.3 Comparative Summary

Direct comparison across the 25-song dataset reveals a consistent GoEmotions advantage. GoEmotions outperformed VADER on 11 songs, while VADER outperformed GoEmotions on only 2 (Sailor Song and Notion), in both cases by a partial margin, since GoEmotions still earned partial credit. The two systems tied on the remaining 12 songs: both were correct on 6 and both failed on 6. The shared successes, including *I Wanna Be Yours*, *I'll Be There For You*, *Sweater Weather*, and *Cheerleader*, are songs with explicit, unambiguous lyrical sentiment, where lexical valence directly encodes emotional register. The shared failures (*Ordinary*, *Lean On*, *Brooklyn Baby*, *Running Up That Hill*, *Teeth*, and *Labour: The Cacophony*) rely on irony, indirection, or intensity that neither system captured. GoEmotions' advantage was most pronounced on emotionally complex songs: *Heat Waves*, *Love Me Not*, *Missing Out*, and *Forever Young* — all cases where VADER's lexical approach failed entirely.

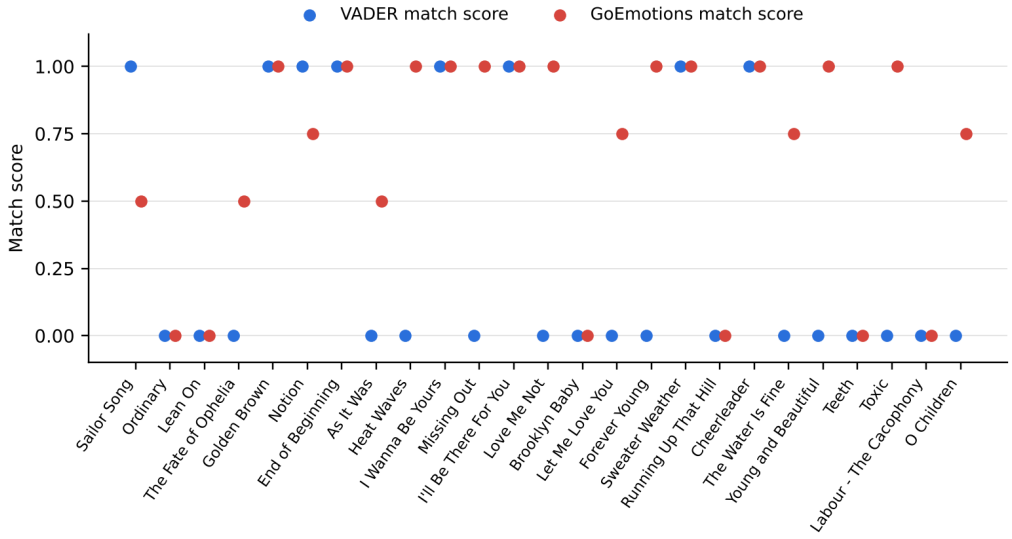


Figure 3: Comparison between VADER Match Scores and GoEmotions Match Scores of Songs

5. Analysis: Failure Taxonomy

5.1 VADER Failure Modes

Qualitative examination of VADER results reveals four recurring failure modes. These failure modes are not unique to the present study — they reflect structural properties of lexical sentiment analysis — but their concentration in the lyrical domain is particularly acute.

Type 1: Metaphor Blindness

Forever Young (Alphaville, VADER: +0.9935, category: Nostalgic) dwells on youth, permanence, and celebration — words that VADER scores as strongly positive. These words, however, carry the song's fear of time passing: the wish to stay young expresses anxiety about aging and loss rather than joy. Their denotative content is celebration, but their affective function is dread. VADER cannot distinguish between words that state a hope and words that betray the fear behind it. This failure mode — the inability to process figurative meaning — has been noted in lexical approaches broadly (Fell & Sporleder, 2014), but lyrics compound it because metaphorical expression is not exceptional but constitutive of the form.

Type 2: Valence Inversion

Ordinary (Alex Warren, VADER: –0.9515, category: Romantic) frames romantic appreciation explicitly through the language of simplicity and mundanity — words VADER scores as negative because they lack conventional positive valence markers. Similarly, *Lean On* (Major Lazer, VADER: –0.9338, category: Nostalgic) expresses dependence and trust through the language of vulnerability and need, which VADER interprets as distress. In both cases, negative-valence vocabulary is deployed in the service of a fundamentally positive emotional register — a rhetorical device common in music but opaque to lexical systems.

Type 3: Affective Collapse

The Bittersweet category produced a within-category VADER range of only 0.124, with *As It Was* (0.798) and *Heat Waves* (0.842) scoring nearly identically and *O Children* (0.922) slightly higher. This apparent consistency conceals a failure: three songs with meaningfully distinct emotional textures — one about estrangement and growth, one about grief and fantasy, one an elegy with eschatological weight — are collapsed into near-identical scores. VADER cannot encode qualitative differences within a broad affective region; it distinguishes only along the negative–positive axis, losing the orthogonal dimensions of emotional complexity, ambivalence, and register.

Type 4: Accidental Accuracy

Sailor Song (Gigi Perez, VADER: –0.9709, category: Longing) was scored as a match: its strongly negative score is consistent with the ache of longing. However, the score derives from the song's nautical imagery — words of distance, absence, and drift — rather than any recognition of what that imagery expresses. These terms function as extended metaphors for deep attachment; their denotative content is loss, but their affective function is love. VADER registered the loss and missed the love, arriving at the right label for the wrong reason. This is a case of correct output through invalid process — what we term accidental accuracy. It represents a distinct epistemological concern: systems that produce correct outputs for wrong reasons will fail unpredictably on related inputs, and their accuracy cannot be taken as evidence of genuine affective understanding.

5.2 GoEmotions Failure Modes

Mode 1: Affective Ambiguity Inflation

GoEmotions frequently predicts neutral as a top-ranked label for songs whose emotional character is specific and identifiable to human listeners. *Running Up That Hill* receives neutral as its highest-ranked label, followed by curiosity and confusion. *Brooklyn Baby* receives neutral, annoyance, and disapproval — a configuration that captures the song's ironic surface register but misses its self-aware romanticism entirely. The model appears to interpret lyrical indirection, irony, or complexity as affective neutrality — an understandable architectural outcome given that GoEmotions was trained on conversational Reddit data, where such registers are rare.

Mode 2: Genre Blindness

Teeth (5 Seconds of Summer, category: Intense) receives love and admiration as dominant GoEmotions labels. The song's second-person address and intimate vocabulary trigger high love probability despite the song's thematic content being conflict, psychological intensity, and destructive attachment. GoEmotions lacks any representation of the semantic context in which love vocabulary appears — a problem of genre: in pop and rock music, intimate address does not reliably index affectionate love. This is a generalization failure specific to musical register.

Mode 3: Emotional Granularity Collapse

Categories such as Empowering, Tragic, and Intense have no direct counterparts in GoEmotions' 28-label taxonomy. *The Water Is Fine* (category: Tragic) receives neutral, love, and caring — labels that partially capture the song's tender emotional openness but entirely miss its tragic valence. GoEmotions has no tragedy label, and its closest analogs — grief, sadness, disappointment — appeared lower in the distribution. This represents a taxonomic gap: human emotional categories for music are not isomorphic to the emotion categories derived from conversational text, and models trained on the latter will systematically underperform on the former.

6. Discussion

The results of this study confirm and extend existing findings on the limitations of text-based sentiment analysis in the musical domain. VADER's 32.0% concordance rate is consistent with the hypothesis that lexical approaches are structurally insufficient for affective classification of music lyrics — not merely imperfect, but architecturally misaligned with the expressive properties of the form. The near-total score bimodality, the within-category score ranges spanning ±1.0, and the four identified failure modes all indicate that VADER is measuring something real (lexical valence) but not what it claims to measure in this context (emotional register).

GoEmotions' substantially higher weighted concordance rate (66.0%) demonstrates that neural multi-label approaches represent a meaningful advance. The richer output format — a ranked distribution over 28 emotion categories — allows partial credit for songs whose emotional

character is captured incompletely but not entirely missed. GoEmotions' superior performance on emotionally complex categories (Longing, Nostalgic, Bittersweet) reflects the value of moving beyond scalar sentiment to categorical emotion prediction.

However, GoEmotions' failure modes reveal that training domain matters as much as architecture. A model trained on Reddit comments carries implicit assumptions about emotional expression that do not transfer cleanly to poetic or musical language: irony is read as neutrality, intimate address is read as affection, and categories that are central to musical experience (tragic, empowering, intense) have no taxonomic home in conversational emotion datasets. Future lyric-specific annotation efforts — analogous to the MoodyLyrics (Çano & Morisio, 2017) or MER datasets — would provide more appropriate training targets for this domain.

These findings have direct implications for music information retrieval and recommendation systems. Streaming platforms increasingly use text-derived features to characterize tracks for mood-based playlist generation. A system relying on VADER to distinguish "sad" from "happy" songs would misclassify the majority of tracks with any lyrical complexity, potentially producing recommendations that feel tonally incoherent to users. GoEmotions-based approaches would perform better but still fail systematically on the most emotionally distinctive tracks — precisely those a user might most want to find. The development of lyric-aware affective models, trained on domain-appropriate annotations and ideally integrating acoustic features, remains an open engineering challenge with significant commercial and artistic implications.

This study is preliminary in scale and scope. A 25-song dataset with single-annotator ground truth does not permit the statistical claims that would be warranted by a larger, multi-annotated corpus. The results should be interpreted as an exploratory demonstration of systematic patterns rather than definitive performance benchmarks. Nonetheless, the consistency and interpretability of the failure modes identified — across two architecturally distinct systems — lends confidence to the conclusion that the domain itself presents challenges that go beyond any particular tool's limitations.

7. Limitations

Several limitations of this study must be acknowledged. First, the dataset of 25 songs was selected by a single annotator, introducing potential selection bias toward emotionally complex or personally meaningful tracks. The overrepresentation of Romantic songs (n=8) may disproportionately influence category-level findings. Second, emotional category assignments and intensity ratings were made without inter-rater reliability verification. Future work should employ multiple independent annotators and compute agreement metrics (e.g., Krippendorff's alpha) to establish ground truth validity.

Third, the dataset is restricted to English-language popular music from predominantly Western cultural contexts, and findings may not generalize across linguistic traditions or musical cultures

with different conventions for emotional expression. Fourth, GoEmotions' graded match scoring was applied by the author using a defined rubric; a more rigorous approach would operationalize match criteria in advance and apply them blindly. Fifth, the present analysis is purely textual and does not incorporate acoustic features, which prior research has established as strong predictors of perceived emotion in music. Any conclusions drawn apply strictly to text-based approaches and should not be interpreted as claims about the limits of computational music emotion recognition broadly.

8. Conclusion

This study demonstrates that both VADER sentiment analysis and GoEmotions multi-label emotion classification fall short of human affective classification when applied to popular music lyrics, achieving binary concordance rates of 32.0% and 48.0% respectively across a 25-song dataset. VADER's failures are structural: its lexical architecture is systematically unable to handle metaphor, valence inversion, affective complexity, and genre register. GoEmotions' failures are domain-specific: its training on conversational text has instilled emotional taxonomies and contextual assumptions that do not transfer to musical language. The two failure profiles are largely complementary, suggesting that combining systems — or training new models on lyric-specific annotated data — could yield meaningful improvements.

The seven failure modes identified — metaphor blindness, valence inversion, affective collapse, and accidental accuracy for VADER; affective ambiguity inflation, genre blindness, and emotional granularity collapse for GoEmotions — constitute a preliminary taxonomy of lyrical sentiment failure that may inform the development of more robust approaches. They point consistently toward the same conclusion: that meaningful computational engagement with the emotional life of music requires more than word frequencies or conversational emotion patterns. It requires, at minimum, acoustic grounding, lyric-specific training data, and affective taxonomies built for the domain.

The gap between what a song says and what it makes us feel remains, for now, beyond the reach of either lexicon or label.

References

Acheampong, F. A., Wenyu, C., & Nunoo-Mensah, H. (2020). Text-based emotion detection: Advances, challenges, and opportunities. *Engineering Reports, 2*(7), e12189. https://doi.org/10.1002/eng2.12189

Çano, E., & Morisio, M. (2017). MoodyLyrics: A sentiment annotated lyrics dataset. *Proceedings of the 2017 International Conference on Intelligent Systems, Metaheuristics & Swarm Intelligence*, 118–124.

Demszky, D., Movshovitz-Attias, D., Ko, J., Cowen, A., Nemade, G., & Ravi, S. (2020). GoEmotions: A dataset of fine-grained emotions. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4040–4054.

Fell, M., & Sporleder, C. (2014). Lyrics-based analysis and classification of music. *Proceedings of COLING 2014*, 620–631.

Hutto, C. J., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the Eighth International Conference on Weblogs and Social Media (ICWSM-14)*.

Kim, Y. E., Schmidt, E. M., Migneco, R., Morton, B. G., Richardson, P., Scott, J., Speck, J. A., & Turnbull, D. (2010). Music emotion recognition: A state of the art review. *Proceedings of ISMIR 2010*, 255–266.

Reagan, A. J., Mitchell, L., Kiley, D., Danforth, C. M., & Dodds, P. S. (2016). The emotional arcs of stories are dominated by six basic shapes. *EPJ Data Science, 5*(1), 31. https://doi.org/10.1140/epjds/s13688-016-0093-1

Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology, 39*(6), 1161–1178.

Appendix A. Full Song Dataset and Classification Results

Song	Artist	Category	Notes	VADER score	VADER match	VADER match score	GoEmotions (top 5 labels)	GoEmotions match	GoEmotions match score
Sailor Song	Gigi Perez	Longing	Deep emotional attachment	-0.9709	Yes	1	curiosity, annoyance, disapproval, anger, love	Partly yes	0.5
Ordinary	Alex Warren	Romantic	Appreciation of simple things	-0.9515	No	0	neutral, desire, annoyance, approval, curiosity	No	0
Lean On	Major Lazer	Nostalgic	Dependence and trust	-0.9338	No	0	neutral, curiosity, confusion, approval, optimism	No	0
The Fate of Ophelia	Taylor Swift	Romantic	Loss and sadness	-0.9081	No	0	neutral, approval, annoyance, caring, sadness	Partly yes	0.5
Golden Brown	The Stranglers	Romantic	Ambiguous emotions	0.0516	Yes	1	neutral, approval, realization, annoyance, disapproval	Yes	1
Notion	The Rare Occasions	Reflective	Questioning reality	0.3453	Yes	1	approval, neutral, disapproval, optimism, confusion	Mostly yes	0.75
End of Beginning	Djo	Nostalgic	Looking back on life	0.5058	Yes	1	neutral, approval, optimism, caring, confusion	Yes	1
As It Was	Harry Styles	Bittersweet	Change and growth	0.7976	No	0	neutral, disapproval, annoyance, confusion, disappointment	Partly yes	0.5
Heat Waves	Glass Animals	Bittersweet	Nostalgia and loss	0.842	No	0	desire, sadness, disappointment, neutral, annoyance	Yes	1
I Wanna Be Yours	Arctic Monkeys	Romantic	Vulnerability and devotion	0.9313	Yes	1	desire, neutral, approval, optimism, love	Yes	1
Missing Out	Maya Hawke	Reflective	Regret and self-discovery	0.9428	No	0	neutral, realization, disappointment, sadness, annoyance	Yes	1
I'll Be There For You	The Rembrandts	Comforting	Friendship and support	0.9778	Yes	1	caring, neutral, approval, joy, optimism	Yes	1
Love Me Not	Ravyn Lenae	Longing	Desire and uncertainty	0.9887	No	0	desire, love, sadness, disappointment, neutral	Yes	1
Brooklyn Baby	Lana Del Rey	Romantic	Identity and self-awareness	0.9891	No	0	neutral, annoyance, disapproval, approval, disappointment	No	0
Let Me Love You	DJ Snake / Justin Bieber	Longing	Support and reassurance	0.9904	No	0	love, disapproval, anger, annoyance, neutral	Mostly yes	0.75
Forever Young	Alphaville	Nostalgic	Fear of time passing	0.9935	No	0	desire, neutral, curiosity, sadness, disappointment	Yes	1

Song	Artist	Category	Notes	VADER score	VADER match	VADER match score	GoEmotions (top 5 labels)	GoEmotions match	GoEmotions match score
Sweater Weather	The Neighbourhood	Romantic	Desire and atmosphere	0.9941	Yes	1	love, neutral, approval, annoyance, desire	Yes	1
Running Up That Hill	Kate Bush	Empowering	Feels like fighting through obstacles	0.9958	No	0	neutral, curiosity, confusion, love, approval	No	0
Cheerleader	OMI	Romantic	Positive and upbeat	0.9964	Yes	1	love, approval, neutral, disapproval, realization	Yes	1
The Water Is Fine	Chloe Ament	Tragic	Emotional openness	0.9966	No	0	neutral, love, approval, realization, caring	Mostly yes	0.75
Young and Beautiful	Lana Del Rey	Romantic	Fear of losing love	0.9977	No	0	love, curiosity, sadness, confusion, caring	Yes	1
Teeth	5 Seconds of Summer	Intense	Conflict and tension	0.9983	No	0	love, admiration, confusion, approval, disappointment	No	0
Toxic	BoyWithUke	Intense		0.001	No	0	sadness, disappointment, annoyance, neutral, disapproval	Yes	1
Labour - The Cacophony	Paris Paloma	Intense		0.8802	No	0	curiosity, annoyance, disappointment, confusion, neutral	No	0
O Children	Nick Cave & The Bad Seeds	Bittersweet		0.9218	No	0	joy, sadness, neutral, disappointment, approval	Mostly yes	0.75

The 25-song dataset with VADER scores, GoEmotions predictions, and match scores used in the study.